

MUCHIRI COLLINS KAMAMA

Solution Architect | Full Stack Developer | Data Scientist

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EXECUTIVE SUMMARY

Versatile technology professional with 5+ years of experience architecting and delivering data-intensive solutions at the intersection of software development, data science, and international development. Combines deep expertise in full-stack development (Node.js, TypeScript, React, Python), advanced database engineering (PostgreSQL HA environments handling 500K+ daily transactions), and applied machine learning (NLP/LLM integration via RAG/LangChain) with firsthand understanding of UN systems, humanitarian operations, and evidence-based programming. Proven track record of building high-availability platforms achieving 99.9% uptime, reducing operational costs by 35%, and enabling data-driven decision-making for organizations including the United Nations, international NGOs, government agencies, and commercial enterprises. Uniquely positioned to bridge technical innovation and development impact translating complex capabilities into accessible solutions for diverse stakeholders.

CORE COMPETENCIES

Full-Stack Development: Node.js | TypeScript | JavaScript (ES6+) | Express.js | React.js | REST APIs | GraphQL | Flask/Django | Next.js

Data Science & Analytics: Python (Pandas, NumPy, scikit-learn) | Statistical Modeling | Predictive Analytics | NLP | LLM/RAG (LangChain) | Hypothesis Testing | Sampling Methods

Database Engineering: PostgreSQL (Advanced: HA, Replication, Tuning, Query Optimization) | MySQL | MongoDB | Redis | ETL Pipelines | Data Warehousing | Backup & DR

Cloud & Infrastructure: AWS (EC2, S3, Lambda, RDS) | Azure DevOps | Docker Containerization | CI/CD | Git/GitHub/GitLab | Linux | IaC Fundamentals

Data Visualization & BI: Power BI (RLS, DAX, Advanced Modeling) | Tableau | Interactive Dashboards | Advanced Excel (VBA, Power Query, Pivot Tables)

Development Sector Expertise: UN System Operations | M&E Methodologies | Donor Reporting (PEPFAR, World Bank) | Research Quality Standards | ODK/KoboToolbox | Geospatial Analysis (ArcGIS)

Professional Skills: Technical Leadership & Mentoring | Stakeholder Management | Cross-Cultural Communication | Agile/Scrum | Project Ownership | Technical Writing | GDPR Compliance

PROFESSIONAL EXPERIENCE

UN Volunteers (UNV) / UNDP Kenya | *Online Volunteer, Monitoring & Evaluation (GEF SGP)* Sep 2026 – Present

- Review grantee narrative reports for completeness, quality, and consistency against approved proposals and workplans for the GEF Small Grants Programme (GEF SGP)
- Support review of grantee financial reports against approved budgets and reporting requirements; track reporting schedules and follow up on outstanding reports
- Contribute to compilation and analysis of project data and results for donor reporting, annual reporting, and terminal evaluation requirements
- Maintain and organize programme databases and electronic filing systems, strengthening records management and audit readiness
- Document key outcomes from workshops, stakeholder meetings, and National Steering Committee engagements, capturing results and lessons learned

Kamama Consulting Solutions | *Principal Technology Consultant & Founder* Jan 2020 – Present (5+ Years)

- Founded and lead technology consulting firm delivering full-stack solutions, database architecture, and data analytics to telecom, NGO, government, and SME clients across East Africa
- Architected and deployed 10+ cloud-native web applications on AWS using Node.js/Express backends with RESTful and GraphQL APIs, reducing client infrastructure costs by 35% while improving scalability
- Designed Tier 3 PostgreSQL high-availability environments with streaming replication, SSL/TLS encryption, query optimization, and automated disaster recovery achieving 99.9% uptime for 500,000+ daily transactions
- Developed full-stack blockchain-based audit management system using Flask/Ethereum backend with React frontend, providing tamper-proof audit trails for procurement workflows and increasing transparency by 35%
- Built predictive analytics models using Python (scikit-learn, Pandas) achieving 85% accuracy on business forecasting; implemented ETL pipelines automating monthly reporting cycles from ~5 days to under 4 hours
- Designed interactive Power BI and Tableau dashboards supporting executive decision-making, donor compliance reporting, and operational performance monitoring across multiple sectors
- Integrated fragmented datasets from 10+ regional projects into centralized databases with standardized schemas, improving cross-regional reporting consistency for international NGO operations

Binti Rising Initiative CBO / SSK Programme | *M&E Officer* Jan 2026 – Present

- Manage field ICT tools and digital data collection systems (ODK/KoboToolbox) for PEPFAR-supported health programme monitoring across four thematic pillars and dispersed regional teams in Kenya
- Develop automated Power BI performance dashboards for real-time milestone tracking and evidence-based decision-making; author comprehensive quarterly M&E reports translating field data into actionable insights
- Apply statistical methods to analyze programme indicators, identify trends, measure outcomes, and support data-driven strategic adjustments for multi-partner consortium accountability
- Coordinate administrative processes maintaining organized documentation systems ensuring donor compliance and audit readiness for international health funding

United Nations Office at Nairobi (UNON) | *Information Management Assistant (Intern)* Jan 2025 – Jul 2025

- Optimized PostgreSQL-supported archival datasets containing 1,200+ records through systematic indexing, validation rules, and structured retrieval workflows, improving data accessibility and query performance by 30%

- ▣ Designed and implemented Python automation scripts (Pandas, NumPy) to ingest, clean, validate, and transform 550+ records, reducing manual processing time by 15% while ensuring data quality standards
- ▣ Coordinated GDPR-compliant data migration of 2,000+ sensitive records across UN office blocks, achieving zero compliance violations through rigorous quality control protocols and comprehensive documentation
- ▣ Maintained SharePoint Online digital filing systems per UN information security policies; trained 55+ staff on Microsoft Teams, SharePoint, and digital records workflows, demonstrating technical communication and mentorship abilities
- ▣ Gained direct familiarity with UN operational protocols, ethics requirements, inter-agency coordination, and organizational culture applicable to all UN system positions

Nakuru County Government | *ICT & E-Government Intern*

May 2022 – Jul 2022

- ▣ Migrated 520+ legacy public-sector financial and administrative records into PostgreSQL database, reducing service delivery turnaround by 30% and improving data accessibility for 100+ county staff
- ▣ Provided on-site technical support across 5 county departments, resolving 150+ hardware, software, and network issues with 95% first-contact resolution rate
- ▣ Implemented endpoint security controls, antivirus deployment, and firewall configurations across 50+ workstations, reducing security incidents by 40% through disciplined risk mitigation
- ▣ Automated financial and administrative reporting processes using Python and Excel automation, reducing report generation time by 25%

Metro Supermarket Group | *Operations Manager & IT Systems Coordinator*

Jun 2019 – Mar 2022

- ▣ Spearheaded transition from manual ledger-based operations to fully digital inventory and POS ecosystem, eliminating reconciliation errors for KES 100,000+ monthly transactions with 99.9% data accuracy
- ▣ Applied inventory analytics and demand forecasting to reduce stock-outs by 25%, improve supplier coordination (15+ suppliers, 95% on-time delivery), and optimize cash flow management
- ▣ Supervised 25+ staff across operations while enforcing financial compliance (zero tax penalties); created Excel-based financial dashboards reducing reporting time by 40%

STRATEGIC TECHNICAL PROJECTS

SDG Knowledge Retrieval System (RAG/NLP/LLM) | Academic Capstone | 2023

Designed and implemented a Retrieval-Augmented Generation system over UN-Habitat SDG 11 documentation using Llama 3.1 and LangChain framework. Built vector embeddings for semantic search, developed serverless inference pipelines on AWS Lambda with GDPR-aligned encryption, and created RESTful APIs for document querying. Reduced information retrieval time by 40%. Demonstrates applied NLP/LLM experience relevant to modern AI-augmented data platforms.

Geospatial Land Restoration & Carbon Reporting Platform | NGO Sector | 2022–2024

Conducted geospatial mapping and spatial analysis of 1,000+ hectares using ArcGIS, integrating GPS coordinates with cloud-hosted PostgreSQL databases. Developed data visualization components for carbon credit tracking and land restoration monitoring dashboards. Applied spatial statistics to support environmental impact assessment for climate-focused programmes.

Blockchain Audit Management System | Enterprise Application | 2024

Developed full-stack Flask/Ethereum application with React frontend providing tamper-proof audit trails for financial and procurement workflows. Implemented REST APIs connecting to existing accounting platforms while aligning with international donor compliance requirements. Demonstrates end-to-end application development capability from database design through deployment.

EDUCATION

Jomo Kenyatta University of Agriculture and Technology (JKUAT)

2019 – 2023

Bachelor of Business Information Technology, Second Class Upper Division Honours

- Relevant Coursework: Database Systems, Software Engineering, Web Development, Business Analytics, Statistics, Research Methods, Data Structures, IT Project Management
- Capstone Project: Predictive analytics model (85% accuracy) using Python scikit-learn and PostgreSQL - Academic Awards: Second Class Upper Division

Jomo Kenyatta University of Agriculture and Technology (JKUAT)

Diploma in Information Technology (2017–2019)

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

- UN Ethics, Information Security, Fraud & Corruption Prevention (2025) - Mandatory UN Training
- Cisco Cybersecurity Essentials Certification (2025)
- Blockchain Fundamentals Certification - Coursera (2025)
- Project Management Essentials (2023)
- In Progress: Google Data Analytics Professional Certificate | AWS Certified Solutions Architect - Associate

ADDITIONAL INFORMATION

Languages: English (Fluent - Professional Working Proficiency) | Kiswahili (Native) | Basic French (Learning)

Work Authorization: Kenyan national; eligible for remote/home-based ICA contracts globally; available for duty stations in Kenya, East Africa, or internationally

Availability: ; comfortable with hybrid remote arrangements, flexible timezone alignment (UTC±3), and international travel (1-4 times/year as required)

Driving: Valid Kenyan Driving License (Classes B, C, C1)

Professional Interests: Evidence-Based Development | Impact Evaluation | SDG Implementation | Data for Social Good | Tech for Humanitarian Action | Open Source Contributions

PROFESSIONAL REFERENCES

- Mrs. Alice Ndungu** - Supervisor, UNON FMTS Unit | Email: gaithuru-ndungu@un.org
Mr. James Ndegwa - Department Administrator, Nakuru County Government ICT | Email: info@nakuru.go.ke | Tel: 051-2214142
Mr. Victor Rotich - Head of Department, IT, PUEA University Nakuru Campus | Tel: 0713-242-910